

The sharp fractional cover frontier at every finite edge/rank ratio

Research continuation, 7 October 2026. We determine the entire limiting fractional-cover curve for intersecting hypergraphs with a bounded edge/rank ratio. The upper bound is an elementary finite theorem using signed counting. Sharpness uses explicitly stated classical design existence and small-intersection covering inputs. All cover values here are fractional unless marked otherwise.

1. Definitions and the finite bound

Let H be a finite simple intersecting hypergraph with $m \geq 1$ nonempty edges of size at most r , where $r \geq 2$ is an integer. Its fractional vertex cover number $\tau^*(H)$ is the minimum sum of nonnegative vertex weights whose sum on every edge is at least one. By finite LP duality, it equals the maximum $Y = \sum_E y_E$ over weights satisfying

$$y_E \geq 0, \quad \sum_{E \ni v} y_E \leq 1 \quad \text{for every vertex } v. \quad (1)$$

Appendix A specifies the finite duality fact used. Every edge is nonempty, so feasible edge weights lie in $[0,1]$ and an optimum exists.

Theorem 1 (finite bound, without additional intersection restrictions). For every integer $k \geq 2$,

$$\tau^*(H) \leq \max \left\{ \frac{(k-1)r+1}{k}, \frac{m}{k+1} \right\}. \quad (2)$$

For $m \geq 2$ we also have $\tau^*(H) \leq m/2$; for $m = 1$ the value is one. No partite, linearity, uniformity, or small-codegree condition is needed. The finite maximum in (2) is not asserted to be the exact optimum for every arithmetic pair (r,m) .

2. A signed bin inequality

Lemma 2. Suppose $k \geq 2$, $1/(k+1) < b \leq 1/k$, and a bin contains $\ell \in \mathbb{Z}_{\geq 0}$ weights in $[0,b]$ of total $W \leq 1 - b$. Then

$$W - (1 - kb)\ell \leq (k-1)\{(k+1)b - 1\}. \quad (3)$$

This is an inequality about the weights assigned to the bin, not all edges incident at the bin's vertex.

Proof. Put $p = (k+1)b - 1 > 0$ and $B = 1 - kb \geq 0$. If $\ell \leq k-1$, use $W \leq \ell b$ to get

$$W - B\ell \leq \ell\{(k+1)b - 1\} \leq (k-1)p.$$

If $\ell \geq k$, use $W \leq 1 - b$ and $B \geq 0$ to get

$$W - B\ell \leq 1 - b - kB = (k^2 - 1)b - (k-1) = (k-1)p.$$

The empty bin belongs to the first case. This proves the lemma. The signed cost is essential: individual bin scores $W - B\ell$ need not be positive. We will sum them only over a genuine partition.

3. Proof of the finite theorem

Fix any feasible vector (1), choose a maximum-weight edge P , and write $b = y_P$, $Y = \sum_E y_E$. Choose any v in P ; feasibility gives $b \leq 1$. Nonnegative weights and intersectingness imply the weighted-star inequality

$$Y - b \leq \sum_{v \in P} \sum_{E \neq P, v \in E} y_E \leq |P|(1 - b) \leq r(1 - b). \quad Y \leq r - (r - 1)b. \quad (4)$$

Multiple intersections do not invalidate this inequality.

Fix k and put $g = (k - 1)r + 1$. If $b \leq 1/(k + 1)$, then $Y \leq m/(k + 1)$, proving (2). If $b > 1/k$, (4) gives $Y \leq r - (r - 1)/k = g/k$, again proving (2).

It remains to consider $1/(k + 1) < b \leq 1/k$. For every edge E other than P , choose **one** vertex $v(E) \in E \cap P$. This is possible by intersectingness. Assign its weight y_E to that vertex's bin. Each edge is assigned exactly once, even if it meets P repeatedly. At a vertex v of P the assigned weights are a subset of the other incident edge weights; therefore their total W_v is at most $1 - b$. Each individual weight is at most b . Let ℓ_v count the assigned edges. We have the exact partition identities

$$\sum_{v \in P} W_v = Y - b, \quad \sum_{v \in P} \ell_v = m - 1. \quad (5)$$

Lemma 2 can now be summed, even though its summands have signs:

$$(Y - b) - (1 - kb)(m - 1) \leq |P|(k - 1)\{(k + 1)b - 1\} \leq r(k - 1)\{(k + 1)b - 1\}. \quad (6)$$

The last factor is positive. Rearranging yields

$$Y \leq m - g + b\{(k + 1)g - km\}. \quad (7)$$

The right side is affine in b . At the endpoints $b = 1/(k + 1)$ and $b = 1/k$ its values are respectively $m/(k + 1)$ and g/k . Therefore (7) is at most their maximum throughout the closed interval. This proves (2) for every feasible vector and hence for the optimum.

For $m \geq 2$, each pair of distinct edges has a common vertex, so $y_E + y_F \leq 1$. Summing over all unordered edge pairs gives $(m - 1)Y \leq \binom{m}{2}$, hence $Y \leq m/2$. For $m = 1$, a unit vertex weight and a unit dual edge weight show $\tau^* = 1$.

This upper-bound proof uses no design theorem, rounding result, numerical experiment or claim in OpenAI/math.

4. The exact limiting curve

Define the continuous nondecreasing function

$$\psi(c) = c/2 \quad (0 \leq c \leq 1), \quad \psi(c) = \max \left\{ \frac{a}{a + 1}, \frac{c}{a + 2} \right\} \quad (a \leq c \leq a + 1, \ a = 1, 2, \dots). \quad (8)$$

The formulas agree at integer endpoints. On $[a, a + 1]$ the switch is $c = a(a + 2)/(a + 1) = a + 1 - 1/(a + 1)$: a plateau is followed by a linear ramp. In particular $\psi(3/2) = 1/2$, $\psi(7/4) = 7/12$, and $\psi(5/2) = 2/3$.

Theorem 3 (sharp frontier at every real ratio). If r_j tends to infinity, H_j is as in Theorem 1, and $m_j/r_j \rightarrow c < \infty$, then

$$\limsup_j \tau^*(H_j)/r_j \leq \psi(c). \quad (9)$$

For every real $c \geq 0$ there is a sequence of **simple intersecting r_j -uniform** hypergraphs with $m_j/r_j \rightarrow c$ for which

$$\tau^*(H_j)/r_j \rightarrow \psi(c). \quad (10)$$

Thus the same sharp limiting curve holds whether uniformity is required or only the rank bound is imposed.

Upper bound. For $c \geq 1$ choose the fixed $k = a + 1$ corresponding to an interval $[a, a + 1]$ containing c , and apply (2). Dividing by r_j and taking limsup gives (9). At shared endpoints either formula works. For $c \leq 1$ use the pair bound, with the single-edge cases contributing at most $1/r_j$. This also handles $c = 0$.

The next section proves (10) and states every imported theorem.

5. Sharpness and classical inputs

A Steiner 2-($v, k, 1$) design is a family of k -point blocks on a v -point set, such that every pair of distinct points belongs to exactly one block. Each point belongs to $r_0 = (v - 1)/(k - 1)$ blocks. For replication at least two, its incidence dual has v distinct edges of size r_0 and is simple and intersecting, and every active vertex has degree k . Uniform dual weights $1/k$ and uniform primal weights $1/r_0$ give matching values v/k , since there are vr_0/k blocks. Thus its fractional cover number is exactly v/k .

Input W (Wilson 1975, fixed-block design existence). For every fixed $k \geq 2$ and all sufficiently large v satisfying

$$k - 1 \mid v - 1, \quad k(k - 1) \mid v(v - 1), \quad (11)$$

a Steiner 2-($v, k, 1$) design exists. In particular one can take an unbounded sequence $v = 1 + k(k - 1)N$, with replication $r_0 = kN$. This classical existence theorem is used only for sharpness, not for the finite upper bound.

Input K (Kahn 1994, Corollary 5.4, printed p.140). For fixed $C > 0$, intersecting r -uniform families with at most Cr edges and maximum distinct-edge intersection $o(r)$ have integer cover number at most $(C/(C + 1) + o(1))r$. Again this is used only in a construction step.

5.1 Plateau construction

Fix $a \geq 1$ and $k = a + 1$. Apply Input W and let H_0 be a design dual with $r_0 \rightarrow \infty$ and $v = (k - 1)r_0 + 1$. It is linear, with $v/r_0 \rightarrow a$. Applying Input K first with $C = a + \eta$ and then decreasing η to zero gives an integer cover C_0 of size

$$|C_0| \leq (a/(a + 1) + o(1))r_0 < r_0 \quad (12)$$

for all sufficiently large r_0 . C_0 is nonempty. All its vertices are original design blocks. The number of original vertices is $n_0 = vr_0/k = \Theta(r_0^2)$.

For any fixed $c \geq a$ choose $L = \max\{0, \lfloor cr_0 \rfloor - v\}$. This is $O(r_0)$. Enlarge C_0 to a fixed set T of $r_0 - 1$ original vertices. For each of L distinct original vertices p outside T , add the edge $T \cup \{p\}$, avoiding edges already in H_0 . There are $n_0 - r_0 + 1$ candidates and at most v forbidden originals, so enough remain. The new edges are distinct r_0 -sets; they meet one another through T and meet every original edge through C_0 . Thus the enlarged hypergraph is simple and intersecting.

Put dual weight $1/k$ on the old edges and zero on the new ones. It remains feasible. Put primal weight $1/r_0$ on every original vertex; every edge, including every new one, has exactly r_0 original

vertices, so its primal sum is one. Both objectives are v/k . Therefore the enlarged family has exactly

$$\tau^* = v/k, \quad m = v + L, m/r_0 \rightarrow c, \quad \tau^*/r_0 \rightarrow a/(a+1). \quad (13)$$

No explicit algorithm to find Wilson's designs or C_0 is asserted; their existence follows from the two named published inputs. For prime-power k , affine-line designs give explicit design instances.

5.2 Linear-ramp construction

For c in $(0,1]$ use the complete pair design on v points (block size two). For c in $[a, a+1]$, $a \geq 1$, use Input W with block size $a+2$. Let r_0 be its replication and set $r = \lceil v/c \rceil$. In both cases $r \geq r_0$. Pad each original edge with $r - r_0$ distinct private vertices, different across edges. Uniformity becomes r ; all original pair intersections remain exactly one, so simplicity and intersectingness are preserved.

Keep uniform dual weights $1/k$, where k is the block size, and primal weights $1/r_0$ on old vertices and zero on the padding. These are feasible with equal value v/k . Hence

$$m/r \rightarrow c, \quad \tau^*/r \rightarrow c/k. \quad (14)$$

This gives $c/2$ below one and $c/(a+2)$ on $[a, a+1]$. Choose between (13) and (14) according to the larger term in (8), proving sharpness. For $c = 0$ take one r -edge with $r \rightarrow \infty$; its value one divided by r tends to zero. Theorem 3 is proved.

5.3 Explicit partite sharpness by shifting a partial pencil

The preceding all-integer construction inputs are unnecessary when the block size is a prime power. We now give an explicit construction that additionally preserves a partition into as many parts as the rank.

Proposition 3.1. Every plateau for which $a+1$ is a prime power, and every ramp for which $a+2$ is a prime power, is attained by simple intersecting partite uniform families with **equal integer and fractional cover numbers**. In particular the entire curve (8) is sharp within partite uniform families for $0 \leq c \leq 4$.

Proof. Fix a prime power q , let $V = \mathbb{F}_q^N$ with $N \geq 3$, and put $v = q^N$, $r_0 = (v-1)/(q-1)$. Vertices of the hypergraph are affine lines; its original edge at $p \in V$ is the set of all lines through p . For each one-dimensional linear direction D , the lines of that direction form one part. There are r_0 parts and each original edge uses exactly one vertex of each part. Any two original edges meet at their unique common line. Every line belongs to q original edges. Uniform old-edge weights $1/q$ have value v/q . All v/q vertices in a fixed part form an integer cover with the same value.

For each two-dimensional linear subspace W choose $z_W \in W \setminus \{0\}$. Define a new edge Q_W by choosing, for each direction D , the line

$$\begin{cases} z_W + D, & D \subset W, \\ D, & D \not\subset W. \end{cases} \quad (14a)$$

This still uses one old vertex per part. These lines cover all points of V : a point in W lies on a line through z_W of a direction inside W ; a point outside W lies on its line through zero of a direction outside W . Thus Q_W meets every original edge. Two such new edges share the zero-line in any direction outside both subspaces. Such a direction exists because

$$r_0 > 2(q+1) \quad (N \geq 3, q \geq 2).$$

Indeed r_0 is at least $q^2 + q + 1$, and $q^2 - q - 1$ is positive. Therefore all new edges are pairwise intersecting.

The new edge changes precisely the q directions in W other than $\langle z_W \rangle$. At least two changed directions span W ; hence different subspaces give different new edges. Each new edge also retains at least two distinct lines through zero, since $r_0 - (q + 1) \geq q^2$. If it were an original point-pencil, that point would be zero, contradicting the changed directions. Thus all old and new edges are distinct.

The number of two-dimensional linear subspaces is

$$\frac{(v-1)(v-q)}{(q^2-1)(q^2-q)} = \Theta(v^2). \quad (14b)$$

This follows by counting ordered independent vector pairs and then ordered bases of a fixed two-dimensional space. For any fixed $c \geq q - 1$, add $L = \max\{0, \lfloor cr_0 \rfloor - v\}$ different new edges. There are more than enough choices since $L = O(r_0)$. Extend the old dual weights by zero. A whole original part remains an integer cover for every new edge, so

$$\tau(H) = \tau^*(H) = v/q, \quad m/r_0 \rightarrow c, \quad \tau^*(H)/r_0 \rightarrow (q-1)/q. \quad (14c)$$

Here τ denotes the ordinary minimum number of vertices meeting every edge. This proves the claimed plateau sharpness.

For a ramp, use just the original affine-line design dual and pad its edges privately to rank $r = \lceil v/c \rceil$, where $0 < c \leq q - 1$. Put the private vertex in position j of every edge into its own new part j , distinct across edges. There is exactly one vertex in each old or new part on every edge. One old part is still an integer cover, and the old dual weights remain feasible. Thus $\tau = \tau^* = v/q$ and $\tau^*/r \rightarrow c/q$. For $q=2$ this covers the interval below one; for $q=a+2$ it gives the ramp in (8). The adjacent block-size pairs (2,3), (3,4) and (4,5) cover all intervals up to $c=4$. At $c=0$ a single edge with singleton parts suffices. This proves the final claim. Finite fields of prime-power order are the standard algebraic input; Wilson and Kahn are not used in this explicit proposition.

These plateau examples permit very large repeated intersections. If $c > q - 1$, the new edges alone give

$$I(H) = \sum_{\{E,F\}} (|E \cap F| - 1) \geq \binom{L}{2} \{r_0 - 2(q+1) - 1\},$$

so $I(H)/r_0^2 \rightarrow \infty$. The sharp fractional result therefore covers families outside any small-intersection regime. These examples supply integer lower constructions; they do not give an integer upper bound for arbitrary families.

6. Exact equality and deletion stability inside the linear phases

Fix an integer $k \geq 2$, put $g = (k-1)r + 1$, and suppose

$$A = km - (k+1)g > 0. \quad (15)$$

Thus the second term in (2) is strictly larger than the first.

Theorem 4 (finite equality). Under (15),

$$\tau^*(H) = m/(k+1) \iff \max_v \deg_H(v) \leq k+1. \quad (16)$$

Proof. Choose an optimal feasible vector. If its maximum b exceeds $1/k$, the weighted-star bound gives a value at most g/k , strictly smaller than $m/(k+1)$. If $1/(k+1) < b \leq 1/k$, (7) rewrites as

$$Y \leq m/(k+1) - A\{b - 1/(k+1)\} < m/(k+1). \quad (17)$$

Thus equality requires b at most $1/(k+1)$. Since m such weights sum to $m/(k+1)$, every weight equals $1/(k+1)$. Vertex feasibility implies the degree cap. Conversely that cap makes the uniform edge weights $1/(k+1)$ feasible, attaining the finite upper bound. This proves (16).

Theorem 5 (explicit deletion bound). Under (15), write $d = m/(k+1) - \tau^*(H) \geq 0$. If

$$d < A/\{k(k+1)\}, \quad (18)$$

there is a set of s edges whose deletion leaves maximum degree at most $k+1$, where

$$s \leq (k+1)(k+2)(1 + m/A)d. \quad (19)$$

The number s is an integer, so the displayed real upper bound also bounds it after rounding down.

Proof. Again use an optimal vector with maximum b . Condition (18) means $Y > g/k$, so b cannot exceed $1/k$. If b exceeds $1/(k+1)$, (17) implies

$$b - 1/(k+1) \leq d/A. \quad (20)$$

Delete precisely the edges with $y_E \leq 1/(k+2)$. Each retained edge has weight strictly above that threshold, so feasibility allows at most $k+1$ retained edges at any vertex. If b is at most $1/(k+1)$, the nonnegative deficits $1/(k+1) - y_E$ sum to d . Each deleted edge contributes at least $1/\{(k+1)(k+2)\}$; hence s is at most $(k+1)(k+2)d$. If b exceeds $1/(k+1)$, sum the nonnegative peak deficits instead:

$$\sum_E (b - y_E) = mb - Y = m\{b - 1/(k+1)\} + d \leq (1 + m/A)d.$$

Each deleted edge again contributes at least $1/\{(k+1)(k+2)\}$. This proves (19) in both cases. Empty cores are allowed. No assumption about zero-weight edges is made.

Corollary 6 (complete stability criterion on a strict ramp). Fix $k \geq 2$ and a real $c \in (k-1/k, k]$. For any sequence as in Theorem 3 with $m_j/r_j \rightarrow c$,

$$\frac{\tau^*(H_j)}{r_j} \rightarrow \frac{c}{k+1} \iff \text{deleting } o(r_j) \text{ edges leaves maximum degree at most } k+1. \quad (21)$$

For the forward direction, $A_j/r_j \rightarrow kc - k^2 + 1 > 0$, while $d_j/r_j \rightarrow 0$. Therefore (18) holds eventually and (19) is $o(r_j)$. For the converse, put weight $1/(k+1)$ on retained edges and zero on all deleted edges. This is feasible on the original family and has value $(m_j - o(r_j))/(k+1)$. The finite upper bound and (8) complete the squeeze. All parameters k, c remain fixed in this limit.

The phase-switch point $c=k-1/k$ is excluded: A_j need not be of linear order there, and (19) does not give a uniform coefficient. The exclusion is essential, already for $k=2$. More generally let k be a prime power and take Proposition 3.1 at $c = k - 1/k$. Then $L/r_0 \rightarrow 1 - 1/k$, and the fractional value is asymptotically sharp. Any $k+2$ new edges retain a common zero-line outside all their selected two-spaces once $r_0 > (k+2)(k+1)$. Therefore any core of maximum degree at most $k+1$ retains at most $k+1$ new edges. It must delete at least $L - (k+1) = \Theta(r_0)$ edges. This disproves the forward implication in (21) if the switch is included.

This corollary does not assert integer-cover recovery. Unlike at integer design ratios, the degree cap alone here need not make intersection excess $o(r_j^2)$. A global rounding theorem requires a separate argument. For an explicit illustration, take the $q = 3$ affine design dual and clone every vertex in one parallel class t times, with each copy assigned to its own new part. Every edge now has rank $r = r_0 + t$, every active degree remains three, and a whole old part still gives $\tau = \tau^* = v/3$. Exactly $v/3$ point triples have their pair intersections increased by t , so $I = tv$. If $t/r_0 \rightarrow 2/c - 1$ for fixed $c \in (3/2, 2)$, then $m/r \rightarrow c$, $\tau^*/r \rightarrow c/3 = \psi(c)$, but $I/r^2 \rightarrow c(1 - c/2) > 0$. This verifies why near-linearity of the retained core cannot simply be inferred on a strict noninteger ramp.

7. Comparison, verification and remaining questions

This replaces the previous repository's nonlinear fractional envelope ϕ and its conservative noninteger gap by the exact asymptotic frontier ψ . The previous frozen package remains a correct historical proof. At integer ratios the new curve agrees with the known design values; at nonintegers it supplies the complete value rather than an unspecified gap. The separate fractional design-stability theorem and its integer recovery corollary are unchanged.

This is not an integer-cover theorem. No unrestricted Ryser solution, journal acceptance, first-result claim or prize-level assessment is made. The finite optimal value at each arithmetic pair (r, m) and a full classification of noninteger extremizers remain open within this project.

The finite upper bound for a feasible edge-weight vector, including the genuine hypergraph-to-bin assignment and all partition sums, is proved in the pinned Lean project and supported by exact rational replay. Formal coverage is recorded separately; it does not include design existence, published rounding, LP duality or all real-limit arguments. Fixed primal/dual certificates illustrate both finite phases, and no solver output is used as evidence for a theorem. The exploratory LP searches helped select the conjectured curve; the proof above is independent of them.

Appendix A. Finite duality

For the finite vertex-by-edge incidence matrix A , weak duality follows by summing the feasibility constraints. A primal vector can be clipped coordinatewise into $[0, 1]$ without destroying feasibility. Every nonempty edge bounds its dual coordinate by one. Thus both extrema exist. Let α be the primal minimum. For $\beta < \alpha$, the point $(\mathbf{1}, \beta)$ lies outside the finitely generated closed cone

$$\{(A^T z - s, \mathbf{1}^T z + t) : z, s, t \geq 0\} \subset \mathbb{R}^{m+1}.$$

Finite-dimensional strict separation gives (u, b) whose value is nonnegative on the cone and negative at this point. The coordinate generators give $u \leq 0$ and $b \geq 0$. Put $y = -u \geq 0$. The vertex generators give $Ay \leq b\mathbf{1}$, while negativity at the separated point gives $\sum_E y_E > b\beta$. If $b = 0$, nonempty edges force $y = 0$, contradicting strict separation. Thus y/b is feasible with value greater than β . Let $\beta \uparrow \alpha$ and use dual compactness. Weak duality then proves equality of the two extrema.

For completeness, the cone is closed as follows. Any representation on a dependent support can lose one generator: move its coefficients along a linear relation until one becomes zero, preserving nonnegativity. Iterate to an independent support. A cone on linearly independent generators is closed because the inverse map on their span is continuous. There are only finitely many such subsets, so the original cone is a finite union of closed cones. The standard finite-dimensional separation theorem is the background input.

References and provenance

[W] R. M. Wilson, *An existence theory for pairwise balanced designs, III: Proof of the existence conjectures*, J. Combin. Theory Ser. A **18** (1975), 71–79, DOI [https://doi.org/10.1016/0097-3165\(75\)90067-9](https://doi.org/10.1016/0097-3165(75)90067-9). The primary abstract states the existence theorem and both divisibility conditions. Its published theorem is an explicitly named construction input.

[K] J. Kahn, *On a Problem of Erdős and Lovász. II: $n(r)=O(r)$* , J. Amer. Math. Soc. **7** (1994), 125–143, Corollary 5.4, DOI <https://doi.org/10.1090/S0894-0347-1994-1224593-5>. The primary Section 5 text was read; the integer covering theorem is prior work, not a contribution of this note.

[FKS] Z. Füredi, J. Kahn and P. D. Seymour, *On the fractional matching polytope of a hypergraph*, Combinatorica **13** (1993), 167–180, DOI <https://doi.org/10.1007/BF01303202>. Its rank-dependent weighted matching results are classical context, not a proof input for (2). The limited comparison does not certify historical novelty.